

Accepted Manuscript

Title: Amputation of congo red dye from waste water using microwave induced grafted *Luffa cylindrica* cellulosic fiber

Author: Vinod Kumar Gupta Deepak Pathania Shilpi Agarwal
Shikha Sharma



PII: S0144-8617(14)00389-0
DOI: <http://dx.doi.org/doi:10.1016/j.carbpol.2014.04.032>
Reference: CARP 8793

To appear in:

Received date: 27-2-2014
Revised date: 18-3-2014
Accepted date: 7-4-2014

Please cite this article as: Gupta, V. K., Pathania, D., Agarwal, S., & Sharma, S., Amputation of congo red dye from waste water using microwave induced grafted *Luffa cylindrica* cellulosic fiber, *Carbohydrate Polymers* (2014), <http://dx.doi.org/10.1016/j.carbpol.2014.04.032>

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

**Amputation of congo red dye from waste water using microwave
induced grafted *Luffa cylindrica* cellulosic fiber**

Vinod Kumar Gupta^{*a}, Deepak Pathania^b, Shilpi Agarwal^a Shikha Sharma^c

^aDepartment of Chemistry, Indian Institute of Technology Roorkee, Roorkee- 247667, India

^bSchool of Chemistry, Shoolini University, Solan -173212, Himachal Pradesh (India)

^cDepartment of Higher Education, Shimla, Himachal Pradesh (India)

^{*}Corresponding author; Email: vinodfcy@gmail.com; dpathania74@gmail.com (D.

Pathania); Fax: 00911332286202; [Tel:00911332285801](tel:00911332285801), +919805440648

ABSTRACT

The present study deals with the surface modification of *Luffa cylindrica* fiber through graft copolymerization of methyl acrylate/acrylamide (MA/AAm) via microwave radiation without the use of initiator. Various reaction parameters effecting grafting yield were optimized and physico-chemical properties were evaluated. The grafted *Luffa cylindrica* fiber showed morphological transformations, thermal stability and chemical resistance. The adsorption potential of modified fiber was investigated using adsorption isotherms for hazardous congo red dye removal from aqueous system. The maximum adsorption capacity of dye onto grafted *Luffa cylindrica* fiber was found to be 17.39 mg/g with best fit for Langmuir adsorption isotherm. The values of thermodynamic parameters such as enthalpy change, ΔH^0 (21.27 kJ/mol), entropy change, ΔS^0 (64.71 J/mol K) and free energy change, ΔG^0 (-139.52 kJ/mol) were also calculated. Adsorption process was found spontaneous and endothermic in nature.

Keywords: *Luffa cylindrica* fiber; graft copolymerization; physico-chemical properties; adsorption capacity; thermodynamics

45 1. Introduction

46 Recently, the surface modification of natural polymers has received great consideration
47 with new developments in science and technology. Lignocellulosic materials developed with
48 substantial alteration in physicochemical properties have been mainly employed as electrical
49 insulators, thermal insulators, vacuum sealants, adsorbents and matrix materials for
50 composites etc. The desirable and targeted physico-chemical properties can be added to
51 natural polymers through various physico-chemical methods in order to meet the specialized
52 applications (Bhattacharya & Misra, 2004; Gupta, Jain, & Varshney, 2007).

53 Polysaccharides of vegetable origin are unique raw materials as they are abundant in
54 nature, inexpensive, biodegradable, renewable, stable, hydrophilic and modifiable
55 biopolymers. The dried fruit of *Luffa cylindrica* (Lc) has been used as good source of
56 lignocellulosic fiber. Luffa sponge contains about 60% of cellulose, 30 % hemicellulose and
57 10% lignin (Rowell, James, & Jeffrey, 2002). The heterogeneity makes the fibers a potential
58 raw material for many industrial applications. However, the high level of moisture
59 absorption, low bulk density, difficulty in dispersion and insufficient adhesion between fibers
60 and polymer matrix are some drawbacks of natural fibers which become critical issue for
61 industrial applications (Zhenping, Xiulin, Mingchen, Chen, & Zhang, 2003; Pathania, &
62 Sharma, 2012). Thus, to improve the compatibility between cellulosic chains and
63 hydrophobic polymer matrices, various physical or chemical surface treatments have been
64 explored (Clasen & Kulicke, 2002; Pathania & Reena, 2012). Graft copolymerization has
65 been considered to be a powerful method for surface modification of lignocellulosic fibers
66 (Gupta, Pathania, Sharma, Agarwal, & Singh, 2013). The improvement in dyeing, printing,
67 chemical resistance, water repelling, fiber strength and abrasion resistance are the some
68 advantages of graft copolymerization (Singha & Rana, 2010).

69 The grafting under the influence of microwave radiation is rapid, efficient, clean, cheap,
70 convenient, energy saving and green method (Gupta, Pathania, Sharma, & Singh, 2013).
71 Microwave heating is an alternative to conventional heating techniques as the microwave
72 energy can easily penetrate and all particles can be heated simultaneously, thus reducing heat
73 transfer problems (Bogdal, Penczek, Pielichowski, & Prociak, 2003; Hou, Wang, & Wu,
74 2008). Recently, microwave radiation has been used in the grafting of monomers onto natural
75 fiber without the use of initiator (Jacob, Chia, & Boey, 1997; Singh, Tiwari, Tripathi, &
76 Sanghi, 2004). In the presence of microwave radiation the oxidative reactions are initiated
77 and free radicals are produced, which leads ultimately to graft reactions.

78 The environment pollution due to toxic compounds discharged from the industrial
79 effluent has been increased with advancement of life. Dyes are the most important
80 constituents among the toxic compounds present in the effluent (Gupta & Ali, 2008; Gupta,
81 Agarwal, Pathania, Kothiyal, & Sharma, 2013; Gupta, Pathania, Agarwal, & Singh, 2012;
82 Gupta, Pathania, Kothiyal, & Sharma, 2013). Due to the toxicity of organic dyes to human
83 health, their removal from water system was of great concern. Many methods have been used
84 for the removal of organic dyes from water (Bhattacharyya & Sharma, 2005, Gupta & Ali,
85 2008, Wang & Zhu, 2007). However, due to disadvantages associated with conventional
86 methods, adsorption process has been used for the removal of organic dye. Adsorption is one
87 of the most effective methods, economically viable, technically feasible and socially
88 acceptable method employed for the treatment of waste water containing dye (Pathania &
89 Sharma, 2012; Gupta & Rastogi, 2009; Pathania, Sharma, Kumar, & Kothiyal, 2014;
90 Rathore, Gupta, Pathania, Sharma, 2014).

91 Cellulosic fibers modified with different monomers have been used as adsorbents for the
92 removal of organic impurities from water system due to of their good selectivity, favorable
93 physicochemical stability, remarkable functionality, enhanced surface area and porosity

(Gadhari, Sanghavi, & Srivastava, 2010). The grafting usually increased the adsorption sites and hence enhance the sorption selectivity of different organic pollutants (Gupta, Agarwal, Singh, & Pathania, 2013; Gupta, Pathania, Sharma, Agarwal, & Singh, 2013).

In view of above facts, the present work deals with the evaluating the viability of using microwave radiations for grafting of binary monomer onto the cellulosic *Luffa Cylindrica* fiber without using initiator. The functional and surface chemistry of the grafted fibers were analyzed. The adsorption capacity of the sample was tested for removal of congo red dye from aqueous system. Moreover, adsorption equilibrium isotherms and thermodynamic studies were also investigated. Thus, efforts have been made to convert this biomass into inexpensive and effective material for industrial purposes.

2. Experimental

2.1. Materials

Methyl acrylate (MA), acrylamide (AAM) and congo red dye were purchased from E. Merck Pvt. Ltd., India. Sodium hydroxide, potassium bromide, ethanol and benzene were received from CHD Ltd., India. Acetone (Rankem Pvt. Ltd., India), nitric acid and dimethyl formamide were obtained from SD Fine Pvt. Ltd., India. All chemicals in this study were used as received.

2.2. Instrumentation

X- ray diffraction (XRD) was carried out on X-ray diffractometer (Bruker D8 Advance). Infra red spectra were recorded on FT-IR spectrophotometer (Perkin Elmer Spectrum 400) using KBr pellets. The morphology study of surface was performed by scanning electron microscope (JEOL, JSM-6610LL, Japan). Perkin Elmer (Pyris Diamond, USA) Thermal Analyzer was used to determine the thermal analysis. The concentrations of dye were determined using UV-Visible spectrophotometer (Systronics 117).

118

119 2.2. Extraction and purification of *Luffa cylindrica* fiber (LcF)

120 *Luffa cylindrica* fibers (LcF) were obtained from dried and ripe fruit collected from the
 121 local fields. The fiber was first extracted from the fruit by soaking in water for 24 h and
 122 washed with 2% detergent solution. The fibers were subjected to soxhlet extraction with
 123 acetone for 12 h in order to remove the impurities. Then the fibers were washed thoroughly
 124 with distilled water.

125 2.3. Mercerization of *Luffa cylindrica* fiber

126 In this process, *Luffa cylindrica* fibers were pre-treated with 5% sodium hydroxide for 30
 127 min to increase its hydrophilicity. The alkali treated fibers were washed thoroughly with
 128 distilled water until the pH of wash water come close to neutral. The fibers were then dried in
 129 oven at 50°C for 12 h.

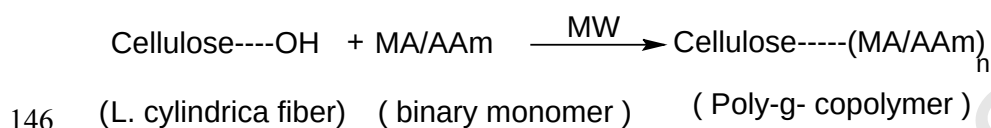


131 2.4. Graft copolymerization of methyl acrylate (MA)/Acrylamide (AAM) onto *Luffa cylindrica*
 132 fiber under microwave irradiation

133 0.5 g of mercerized *Luffa cylindrica* fiber was immersed in 100 mL double distilled
 134 water for 24 hours prior to graft copolymerization in order to activate the reaction sites on the
 135 fiber surface. Then the known amount of methyl acrylate/acryl amide binary monomer in
 136 definite ratio was added with constant stirring for definite time. Different reaction parameters
 137 such as monomer concentration, temperature and microwave exposure time were optimized.
 138 The homopolymer formed during the graft copolymerization was removed with hot distilled
 139 water followed by methanol. The grafted *Luffa cylindrica* fibers (*Lc-g-poly(MA/AAM)*) thus
 140 obtained were dried at 50°C for 12 h in a hot air oven. The percentage grafting yield (G%) of
 141 grafted sample was calculated as follow:

$$\text{Grafting Yield} = \frac{W_3 - W_1}{W_1} \times 100 \quad (1)$$

where, W_1 is the initial weight of the raw fiber and W_3 is the final weight of the grafted fiber after extraction of homopolymer. The general grafting reaction of binary monomer onto *Luffa cylindrica* cellulosic fiber has been shown as follow:



2.5. Characterization of grafted fiber

2.5.1. Fourier transform infrared (FTIR)

Fourier transform infrared spectra of raw and grafted samples were recorded by Perkin-Elmer FTIR spectrophotometer (model 400, USA) using KBr pellets. FTIR spectra of the sample were analyzed in the range of 400-4000 cm^{-1} .

2.5.2. Thermal analysis (TGA/DTA)

Thermo gravimetric and differential thermal analysis of raw and grafted samples were carried out in nitrogen atmosphere at a heating rate of 10°C/minute using Perkin Elmer (Pyris Diamond, USA) thermal analyzer.

2.5.3. X-ray diffraction studies (XRD)

X-ray diffraction studies of raw and grafted sample were performed on Bruker D8 Advanced X-ray diffractometer, using Nickel-filtered Cu $K\alpha$ radiation ($\lambda=0.15406$ nm) and scanned from 2 to 60°C at a scan speed of 20/min. Finely powdered samples of raw and grafted fibers were used to study the crystallinity of the samples.

2.5.4. Morphological studies

Morphological analysis of *LcF* and *Lc-g-poly(MA/AAm)* were carried out on scanning electron microscope (JEOL JSM-6100, Japan).

2.6. Physico-chemical behavior

2.6.1. Swelling behavior

The swelling studies of *LcF* and *Lc-g-poly(MA/AAm)* were performed in water, dimethyl formamide (DMF) and benzene. In this, 0.5 g sample was immersed in 100 mL solvent for 24 h. The excess of solvent was removed with filter paper. The final weight of sample was noted and percentage swelling was calculated by following formula:

$$\% \text{ swelling} = \frac{W_f - W_i}{W_i} \times 100 \quad (2)$$

where W_i is the initial weight of the fiber and W_f is the weight after swelling of fiber.

2.6.2. Water uptake study

The water uptake capacity of *LcF* and *Lc-g-poly(MA/AAm)* was studied using concept of capillary action. In this method, the wicks of different sample of same diameter were prepared and initial ink mark is drawn at one end. These wicks were dipped into beakers containing water for 24 h. The rise in water level in each wick was noted with the help of the scale.

2.6.3. Moisture absorbance studies

0.5 g dry weight of *LcF* and *Lc-g-poly(MA/AAm)* were placed in the humidity chamber for 2 h under 40% humidity level. The final weights of the samples were noted after taking out the sample immediately. The percentage moisture absorbance (M_{abs}) was calculated by following formula:

$$\% M_{abs} = \frac{W_2 - W_1}{W_1} \times 100 \quad (3)$$

where W_1 is the initial weight of the fiber and W_2 is the weight after moisture absorbance.

2.6.4. Chemical resistance studies

Chemical resistance of samples was determined in term of percentage weight loss. In this, known weight of *LcF* and *Lc-g-poly(MA/AAm)* samples were immersed in aqueous

solution of 1N NaOH and 1N HNO₃. Then samples were taken out and dried in hot air oven to constant weight. The final weights of the samples were noted and percentage weight loss was calculated by following formula:

$$\% \text{ Wt loss} = \frac{W_1 - W_2}{W_1} \times 100 \quad (4)$$

where W₁ is the initial weight of the sample and W₂ is the weight after action of acid and base.

2.7. Dye adsorption experiments

The adsorption of congo red (CR) onto grafted sample was performed using batch experiments. In this process, 0.2 g of adsorbent was added to 100 mL of dye solution of different concentrations (50–500 mg/L) and the mixture was agitated in a thermoshaker at a speed of 100 rpm for a given time at 30°C. The suspensions were centrifuged equilibrium concentration of dye in the supernatant liquor was analyzed by double beam UV-visible spectrophotometer (Gupta, Agarwal, Singh, & Pathania, 2013). The experiment conditions were optimized at different concentration, temperature, pH, adsorbent amount and contact time. The amount of dye adsorbed per unit mass of adsorbent, q_e, (mg/g) was obtained using following equations:

$$q_e = \frac{C_0 - C_e}{W} \times V \quad (5)$$

Where q_e (mg/L) is amount of dye adsorbed, C₀ is the initial concentration of congo red solution (mg/L), C_e is the concentration at equilibrium, V is the volume of dye solution and W is the weight of the grafted fibers. The study of isotherms was carried out by varying the concentration of CR dye (50–500 mg/L), volume (100 mL), adsorbent dose (0.50 g), pH (7), time interval (120 min) and temperature (30°C). For thermodynamics studies, observations were made under optimized conditions at different temperatures (20°C -50°C).

3. Results and discussion

3.1. Graft copolymerization of methyl acrylate/acrylamide (MA/AAm) onto *Luffa cylindrica* fiber

The major component of the *Luffa cylindrica* fiber is cellulose. The presence of active hydroxyl group at positions C₂, C₃ and C₄ of cellulose are responsible for graft copolymerization. Before the treatment of fiber with binary monomers it was mercerized with 5% sodium hydroxide to increase the hydrophilicity.

3.1.1. Optimization of grafting parameters

3.1.1.1. Optimization of MA concentration in MA/AAm binary monomer

The results of effect of MA concentration on grafting yields were shown in Table 1. It has been observed the grafting yield decreased with increase in the concentration of MA. It was due to dominance of homopolymerization onto cellulosic fiber with increase in monomer concentration. The maximum grafting yield (49%) was observed at 1.12×10^{-3} mol/L concentration of MA in MA/AAm binary monomer by keeping AAm concentration constant.

3.1.1.2. Optimization of AAm concentration in MA/AAm binary monomer

The effect of AAm concentration on grafting yields was shown in Table 1. It has been observed that grafting yield initially increased with increase in the concentration of AAm and then decreased further with increase in the concentration. The maximum grafting yield (51.08 %) was observed at 2.81×10^{-3} mol/L concentration of AAm in binary monomer by keeping MA concentration constant.

3.1.1.3. Effect of microwave exposure time

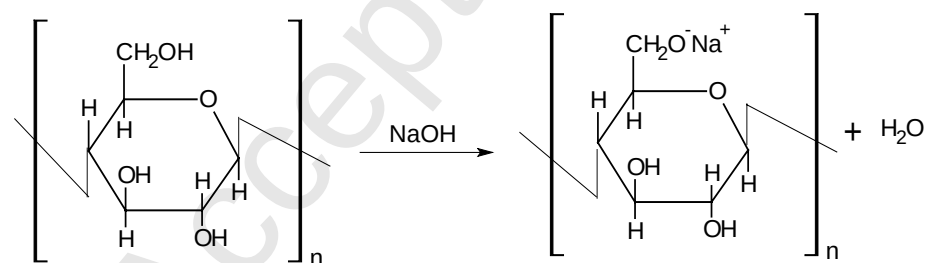
Table 1 shows the variation of grafting yield with time. It is evident that with increase in the microwave irradiation time up to 2 min the grafting yield increases to 48%. Further,

increase in reaction time resulted in decrease in grafting yield due to homopolymer formation and may be degradation of the cellulosic backbone of fiber (Mishra & Sen, 2011; Singh, Tiwari, Pandey, & Singh, 2006).

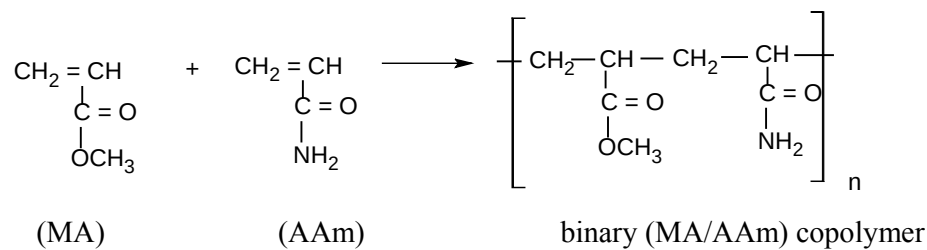
3.2. Mechanism of grafting of MA/AAm on to raw fiber of *Luffa cylindrical*

Cellulose is a large molecule with many –OH groups attached at different positions. It has been observed that in microwave region the resulting dielectric of cellulose molecule may cause an increase in the rate of reaction at these groups. The dielectric heating results rapid energy transfer from these groups to neighboring molecules (MA, AAm and water) as it is impossible to store the energy in specific part of the molecule. The dielectric heating results in bond breaking and creating radical sites at oxygen. Moreover, microwaves also resulted in lowering of Gibbs energy of activation of the reaction (Galema, 1997; Ibrahim, Shuy, Ang, & Wang, 2010). The proposed steps for the mechanism of grafting reactions are as follow (Singha, & Rana, 2010; Mishra, & Sen, 2011; Singh, Tiwari, Tripathi, & Sanghi, 2004):

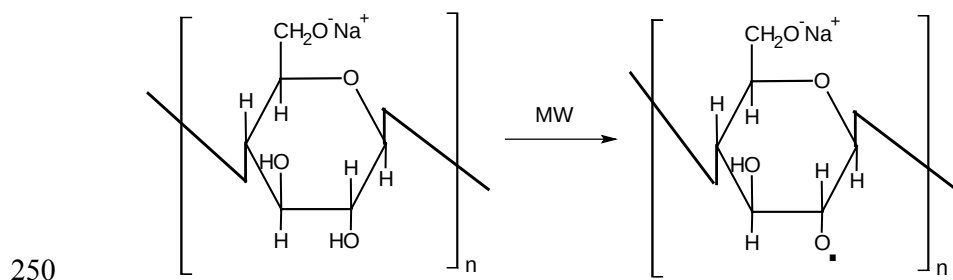
Step 1: Mercerization



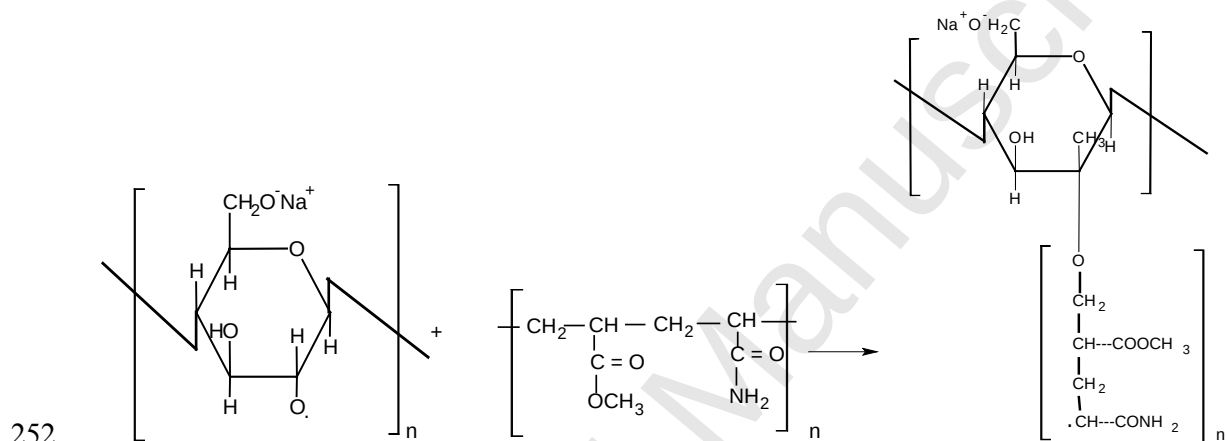
Step 2: Formation of binary monomer



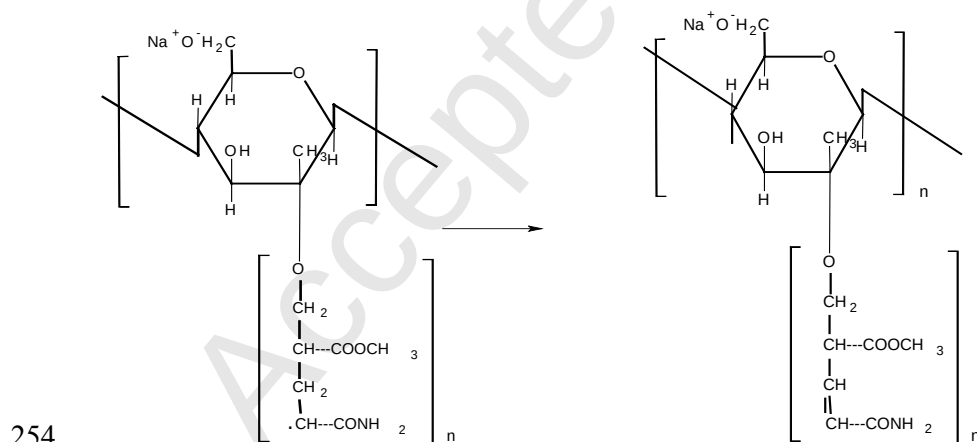
Step 3: Initiation



251 Step 4: Propagation



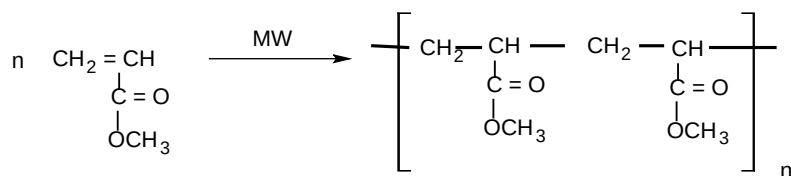
253 Step 5: Termination



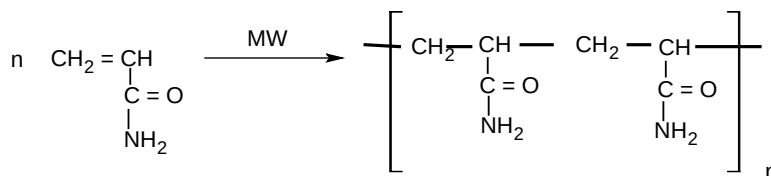
255

256 Step 6: Formation of homopolymers

257 (i) Homopolymer of MA



(ii) Homopolymer of AAm



3.3. Characterization of LcF and Lc-g-poly(MA/AAm)

3.3.1. X-ray diffraction (XRD)

The XRD pattern of LcF and Lc-g-poly(MA/AAm) were shown in Fig. 1(a-b). The LcF at 2θ scale showed peaks at 22.35° and 15.48° with relative intensities of 779 and 442, respectively. Similarly, Lc-g-poly(MA/AAm) showed peaks at 24.23° and 16.40° with relative intensities of 546 and 352, respectively. The percentage crystallinity (X_c %) and crystallinity index (C.I) was calculated as follow (Kalia, Kumar, & Kaith, 2010; Sanghavi, Mobina, Mathur, Lahiri, & Srivastava, 2013):

$$X_c \% = \left\{ \frac{I_c}{I_A + I_c} \right\} \times 100\% \quad (6)$$

$$C.I. = \frac{I_c - I_A}{I_c} \quad (7)$$

where I_c is peak intensity of crystalline phase, I_A is peak intensity of amorphous phase.

The percentage crystallinity of LcF and Lc-g-poly(MA/AAm) fiber was observed as 63.80 and 60.80, while the crystallinity index as 0.43 and 0.35. It was observed that the intensity of the peak in Lc-g-poly(MA/AAm) decreased on grafting. The decrease in intensity of peak during grafting indicated decreased crystallinity of Lc-g-poly(MA/AAm). However, the Lc-g-

poly (MA/AAm) showed broadening of the peak after grafting due to convergence of the fibers towards more disordered system (Sanghavi, Kalambate, Karna, & Srivastava, 2014).

It has been observed that (Table 2) a slight decreased in percentage crystallinity of the fiber on graft copolymerization resulted in increase in randomness or disorder in the crystal lattice of cellulose fiber. This was due to incorporation of *poly(MA/AAm)* chains on the active sites of backbone during grafting and fibers became more amorphous and resulted in impaired crystalline structure (Wang, Dong, & Xu, 2006; Sharma, Pathania, & Singh, 2013).

3.3.2. Fourier transform infra red spectroscopy (FTIR)

The IR spectra of *LcF* and *Lc-g-poly(MA/AAm)* were shown in Fig. 1(c-d). The peak at 899 cm^{-1} may be due to C-C stretching vibration and β -glycosidic linkage (Kaur, Kumar, & Sharma, 2010). The peaks at 2856 cm^{-1} and 2925 cm^{-1} were due to symmetric and asymmetric stretching of C-H bond of $-\text{CH}_2$ (Mishra & Sen, 2011). The broad peak at 3401 cm^{-1} was due to stretching vibration of $-\text{OH}$ (Singh, Tiwari, Tripathi, & Sanghi, 2004). Peaks observed at 1376 cm^{-1} , 1427 cm^{-1} and 1456 cm^{-1} may be due to $-\text{CH}$, $-\text{CH}_2$, and $-\text{CH}_3$ bending, respectively (Singha & Rana, 2010). The additional peaks for grafted sample at 1052 cm^{-1} , 1505 cm^{-1} , 3400 cm^{-1} and 1111 cm^{-1} were due to C-O-H deformation of raw fiber, C-O of amide group of acryl amide, N-H stretching of amide group and C-O groups, respectively (Wan et al., 2011). A sharp peak at 1736 cm^{-1} was observed due to C=O group of ester (methylacrylate) (Kalia, Kumar, & Kaith, 2010; Oei, Ibrahim, Wang, & Ang, 2009). The additional peaks observed in the grafted sample confirmed the grafting of MA/AAm binary monomer onto *Luffa cylindrica* fiber.

3.3.3. Thermal analysis (TGA/DTA)

Thermogravimetric analysis of *LcF* and *Lc-g-poly(MA/AAm)* was carried out as a function of weight loss verses temperature as shown in Fig. 2(a-b). The degradation may

occur in various forms such as deacetylation, dehydration, decarboxylation and chain scissions. In TGA curve of *LcF*, two stage decomposition was observed, with maximum weight loss of 59.5% between 86.1 °C to 332.5 °C in first stage and maximum weight loss of 30.8% between 332.5 °C to 503.2 °C in second stage of decomposition. The first stage decomposition was due to loss of moisture and second stage decomposition was due to cellulosic and lignin degradation (Kalia, Kumar, & Kaith, 2010; Ibrahim, Fatimah, Ang, & Wang, 2010).

The TGA curve of *Lc-g-poly(MA/AAm)* also showed two stage decomposition. The first stage decomposition was observed with 15.7% weight loss between 49.8 °C to 277.6 °C and second stage decomposition with 73.2% weight loss between 277.6 °C to 363.2 °C. This was attributed due to the strengthening of fibers due to increase in covalent bonds in the *Lc-g-poly(MA/AAm)* (Tiwari & Singh, 2008; Pathania, Kumar, & Bhatt, 2009).

Differential thermogravimetric (DTG) curve indicated the decomposition peak at 319.8 °C for *LcF* and 334.7 °C for *Lc-g-poly(MA/AAm)*. Thus DTG studies confirmed the improved thermal resistance of grafted copolymer due to incorporation of covalent bonding through inclusion of *poly(MA/AAm)* (Sanghavi et. al., 2013).

3.3.4. Scanning electron microscopy (SEM)

Fig. 3(a-b) shows the SEM micrographs of *LcF* and *Lc-g-poly(MA/AAm)*. It was revealed that the grafting results in change in morphology of the fibers. The surface of the fibers became rough due to attachment of polymer chains to the surface (Kalia, Kumar, & Kaith, 2010; Pathania & Sharma, 2011).

3.4. Physico-chemical behaviors of *Lc-g-poly(MA/AAm)*

3.4.1. Swelling behavior in different solvents

The swelling behavior of *LcF* and *Lc-g-poly(MA/AAm)* in different solvents (water, DMF and benzene) was shown in Fig. 4(a-b). The *LcF* shows maximum swelling in water and the order of percentage swelling as: water > DMF > benzene. This was due to more affinity of water for free hydroxyl groups in raw cellulosic fibers (Taylor, Fanta, Doane, & Russell, 1978). The *Lc-g-poly(MA/AAm)* copolymer show more swelling in benzene than water and follows the trend as: benzene > DMF > water. This was due to blocking of active sites of *Luffa cylindrica* cellulosic fiber by grafting.

3.4.2. Moisture absorbance studies

The moisture absorbance studies on *LcF* and *Lc-g-poly(MA/AAm)* were carried out under different humidity levels. The percentage of moisture absorbance (M_{abs}) of the samples was shown in Table 3(a). It was observed that the *LcF* have high % M_{abs} (48%) due to presence of hydrophilic hydroxyl group than the % M_{abs} (34%) for *Lc-g-poly(MA/AAm)*. It was due to the blockage of active sites by graft copolymerization which led to decrease in hydrophilic character (Singha & Rana, 2010; Sanghavi, & Srivastava, 2013).

3.4.3. Water uptake studies

The water uptake capacity of *LcF* and *Lc-g-poly(MA/AAm)* was studied using the concept of capillary action. Table 3(b) shows that low water uptake capacity of *Lc-g-poly(MA/AAm)* was due to blocking of active hydrophilic sites of the fiber by graft copolymerization thus decreasing hydrophilicity (Kalia, Kumar, & Kaith, 2010).

3.4.4. Chemical resistance studies

The chemical resistance of *LcF* and *Lc-g-poly(MA/AAm)* was determined in 1N NaOH and 1N HNO₃. It has been observed that *Lc-g-poly(MA/AAm)* was chemically more resistant than *LcF* (Table 3(c)). This was due to deactivation of active sites by graft copolymerization

onto cellulosic chains of the fiber (Pathania, Kumar, & Bhatt, 2009; Sanghavi, & Srivastava, 2011).

3.4.5. Dye adsorption studies

The *Lc-g-poly(MA/AAm)* has been used for removal of congo red dye from water system. The results of dye adsorption onto grafted sample were shown in Fig. 4(c). It has been observed that dye adsorption capacity increased with the increase in concentration of dye. The adsorption of dye onto *Lc-g-poly(MA/AAm)* may be due to the grafting of carboxylic, amide, hydroxyl and ester groups of binary monomers onto cellulosic chains of *Luffa cylindrica* fibers.

3.4.6. Adsorption Isotherms

The adsorption data obtained from experiments provides estimation of maximum adsorption capacity of the adsorbent and effectiveness of adsorbate-adsorbent system. The adsorption capacity and other parameters were evaluated using Langmuir and Freundlich, isotherm models.

The model expressions and their linearized forms are given in Table 4. Linear curve fitting procedure was used to fit the experimental data to the models and to the determination of the model parameters. The values of the isotherms parameters are given in Table 4. The Langmuir isotherm confirmed the monolayer adsorption onto a surface containing a finite number of adsorption sites via uniform strategies of adsorption with no transmigration of the adsorbate taking place along the plane of the surface (Gupta, Pathania, Agarwal, & Sharma, 2012). Fig. 4(d) shows a Langmuir isotherm from which isotherm constants, q_m (monolayer adsorption capacity of the adsorbent, mg/g), and K_L (Langmuir adsorption constant (L/mg), related with the free energy of adsorption) were calculated. It has been observed that the

maximum adsorption capacity (q_m) was found to be 17.39 mg/g. A high value of coefficient of regression, R^2 (0.995) indicated the applicability of Langmuir isotherm. The K_L value determined was further used to calculate the dimensionless separation factor (R_L), which is given as:

$$R_L = \frac{1}{1 + K_L C_e} \quad (8)$$

where, C_e is the equilibrium dye concentration. The magnitude of R_L value gives an idea about the nature of adsorption equilibrium. The $R_L < 1$ (0.427) indicated spontaneous adsorption of dye from aqueous solution (Pathania & Sharma, 2012).

The Freundlich isotherm has been commonly used to describe adsorption characteristics for heterogeneous surface (Pathania, Sharma, & Singh, 2013). Fig. 7(e) shows a Freundlich isotherm from which isotherm constants K_F and n were calculated. The value of $n > 1$ observed from Freundlich isotherm indicated favourable and heterogeneous adsorption. The isotherm constants and coefficient of regression R^2 have been given in Table 4. A comparison of the coefficient of regression (R^2) for the isotherms (Table 4) indicated that the equilibrium data was best fitted in the Langmuir isotherm.

3.4.4. Adsorption thermodynamics

In order to study the thermodynamics of adsorption of congo red dye onto grafted fibers, three basic thermodynamic parameters such as free energy change (ΔG^0), enthalpy change (ΔH^0) and entropy change (ΔS^0) of sorption were calculated using following equations:

$$\ln K_D = -\frac{\Delta G^0}{RT} \quad (9)$$

$$K_D = \frac{C_0 - C_e}{C_e} \quad (10)$$

$$\Delta G^0 = -2.303RT \log K_D \quad (11)$$

The other thermodynamic parameters such as change in standard enthalpy (ΔH^0) and standard entropy (ΔS^0) were determined using the following equation

$$\Delta G^0 = \Delta H^0 - T\Delta S^0 \quad (12)$$

$$\ln K_D = \frac{\Delta S^0}{R} - \frac{\Delta H^0}{RT} \quad (13)$$

where, R is universal gas constant (8.314 kJ/mol K), C_0 and C_e is the initial and equilibrium concentration (mg/L). ΔS^0 and ΔH^0 are obtained from the slope and intercept of the Vant Hoff's plot of $\ln K_D$ versus $1/T$. The positive value of ΔH^0 (21.27 kJ/mol) indicated that dye adsorption was physical and endothermic reaction. An adsorption process is generally considered as physical if $\Delta H^0 < 25$ kJ/mol and as chemical when $\Delta H^0 > 40$ kJ/mol (Gupta et al., 2012). The negative values of ΔG^0 (-139.52 kJ/mol) indicated spontaneous adsorption. Further, the positive value of entropy change, ΔS^0 (64.71 J/mol K) reflected the increased randomness at the solid–solution interface during the fixation of the adsorbate on the active sites of the adsorbent. This may be due to the fact that before the adsorption process starts, the adsorbate ions in solution are heavily solvated and the system is more ordered and the order is lost when the dye species are adsorbed on the surface due to the release of solvated water molecules (Gupta, Pathania, Agarwal, & Sharma, 2013). Moreover, adsorbed solvent (water) molecules which are displaced by the dye species, gain more translation entropy.

4. Conclusions

Graft copolymerization under the influence microwave radiations is one of the best methods for modifying the properties of natural fibers. *Luffa cylindrica* fiber has been successfully grafted without initiator by binary vinyl monomers MA/AAm and MA/AA under microwave radiations. The raw and grafted fiber were characterized by different techniques such as FTIR, XRD, TGA and SEM. FTIR results showed the formation of new bonds on the grafted sample. TGA spectra show thermal stability, XRD spectra revealed the

decrease in the crystallinity after grafting and SEM images confirmed the improved surface morphology. Thus, it is concluded that grafted *Luffa cylindrica* fiber shows improved thermal, structural, chemical, and morphological properties so it can be used as an adsorbent for water purification, reinforcing material in polymer composites and for other industrial applications.

References

- Bhattacharya, A., & Misra, B. N. (2004). Grafting: a versatile means to modify polymers Techniques, factor and applications. *Progress in polymer science*, 29, 767-814.
- Bhattacharyya, K. G., & Sharma, A. (2005). Kinetics and thermodynamics of Methylene Blue adsorption on Neem (*Azadirachta indica*) leaf powder. *Dyes and Pigments*, 65, 51–59.
- Bogdal, D., Penczek, P., Pielichowski, J., & Prociak, A. (2003). Microwave Assisted Synthesis, Cross linking and Processing of Polymeric Materials. *Advance Polymer Science*, 163, 193-263.
- Clasen, C., & Kulicke, W. M. (2001). Determination of viscoelastic and rheo-optical material functions of water-soluble cellulose derivatives. *Proggress in polymer science*, 26, 1839-1919.
- Gadhari, N. S., Sanghavi, B. J., & Srivastava, A. K. (2010). Potentiometric stripping analysis of antimony based on carbon paste electrode modified with hexathia crown ether and rice husk. *Analytica Chimica Acta*, 703(1), 31-40.
- Galema, S. A. (1997). Microwave chemistry. *Chemical Society Review*, 26, 233-238.
- Gupta, V. K., Jain, R., & Varshney, S. (2007). Removal of reactofix golden yellow 3 RFN from aqueous solution using wheat husk - an agricultural waste. *Journal of Hazardous Material*, 142, 443–448.
- Gupta, V. K., Agarwal, S., Singh, P., & Pathania, D. (2013). Acrylic acid grafted *Luffa cylindrica* fiber for the removal of dye and metal ions. *Carbohydrate Polymer*, 98, 1214-1221

- 444 Gupta, V. K., Agarwal, S., Pathania, D., Kothiyal, N. C., & Sharma, G. (2013). Use of pectin-
 445 thorium (IV) tungstomolybdate nanocomposite for photocatalytic degradation of
 446 methylene blue. *Carbohydrate Polymer*, 96, 277-283.
- 447 Gupta, V. K., & Ali, I. (2008). Removal of endosulfan and methoxychlor from water on
 448 carbon slurry. *Journal of Environmental Science & Technology*, 42, 766-770.
- 449 Gupta, V. K., Pathania, D., Agarwal, S., & Sharma, S. (2012). Decolorization of hazardous
 450 dye from water system using chemical modified *Ficus carica* adsorbent. *Journal of*
 451 *Molecular Liquids*, 174, 86-94.
- 452 Gupta, V. K., Pathania, D., Agarwal, S., & Sharma, S. (2013). Removal of Cr (VI) onto *Ficus*
 453 *carica* biosorbent from water. *Journal of Environmental Science and Pollution*
 454 *Research*. 20, 2632-2644.
- 455 Gupta, V. K., Pathania, D., Agarwal, S., & Singh, P. (2012). Adsorptional photocatalytic
 456 degradation of methylene blue onto pectin-CuS nanocomposite under solar light. *J*
 457 *Hazd. Mater.*, 243, 179-186.
- 458 Gupta, V. K., Pathania, D., Kothiyal, N. C., & Sharma, G. (2013). Use of Pectin - thorium
 459 (IV) tungstomolybdate nanocomposite for photocatalytic degradation of methylene
 460 blue. *Carbohydrate Polymer*, 96, 277-287.
- 461 Gupta, V. K., Pathania, D., Sharma, S., Agarwal, S., & Singh, P. (2013). Remediation and
 462 recovery of methyl orange from aqueous solution onto acrylic acid grafted *Ficus carica*
 463 fiber: Isotherms, Kinetics and thermodynamics. *Journal of Molecular Liquids*, 177,
 464 325-334.
- 465 Gupta, V. K., Pathania, D., Sharma, S., & Singh, P. (2013). Preparation of bio-based porous
 466 carbon by microwave assisted H₃PO₄ activation and its use for adsorption of Cr (VI).
 467 *Journal of Colloid and Interface Science*, 401, 125-132.
- 468 Hou, A., Wang, X., & Wu, L. (2008). Effect of microwave irradiation on the physical
 469 properties and morphological structure of cotton cellulose. *Carbohydrate Polymers*, 74,
 470 934-937.
- 471 Ibrahim, S., Fatimah, I., Ang, H., & Wang, S. (2010). Adsorption of anionic dyes in aqueous
 472 solution using chemically modified barley straw. *Water Science Technology*, 62, 1177-
 473 1182.
- 474 Ibrahim, S., Shuy, W. Z., Ang, H., & Wang, S. (2010). Preparation of bioadsorbents for
 475 effective adsorption of a reactive dye in aqueous solution. *Asia-Pacific Journal of*
 476 *Chemical Engineering*, 5, 563-569.

- 477 Jacob, J., Chia, L. H. L., & Boey, F. Y. C. (1997). Microwave polymerization of poly
478 (methyl acrylate): Conversion studies at variable power. *Journal of Applied Polymer*
479 *Science*, 63, 787-790.
- 480 Kalia, S., Kumar, A., & Kaith, B. S. (2010). Sunn hemp cellulose graft copolymers
481 polyhydroxybutyrate composites: morphological and mechanical studies. *Advanced*
482 *Material Letters*, 2, 17-25.
- 483 Kaur, I., Kumar, R., & Sharma, N. (2010). A comparative study on the graft
484 copolymerization of acrylic acid onto rayon fiber by a ceric ion redox system and γ -
485 radiation method. *Carbohydrate Research*, 345, 2164-2167.
- 486 Mishra, S., & Sen, G. (2011). Microwave initiated synthesis of polymethylmethacrylate
487 grafted guar (GG-g-PMMA), characterizations and applications. *International Journal*
488 *of Biological Macromolecules*, 48, 688-694.
- 489 Oei, B. C., Ibrahim, S., Wang, S., & Ang, H. M. (2009). Surfactant modified barley straw for
490 removal of acid and reactive dyes from aqueous solution. *Bioresources Technology*,
491 100, 4292-4295.
- 492 Pathania, D., Kumar, M., & Bhatt, S. S. (2009). Thermal analysis of Caryota Urens fiber
493 grafted with acrylonitrile and methyl methacrylate. *Trends of Carbohydrate Research*,
494 4, 30-33.
- 495 Pathania, D., Sharma, G., Kumar, A., & Kothiyal, N. C. (2014). Fabrication of
496 nanocomposite polyaniline zirconium(IV) silicophosphate for photocatalytic and
497 antimicrobial activity. *Journal of Alloys and Compounds*, 588, 668-675.
- 498 Pathania, D., & Sharma, S. (2011). Characterization of *Ficus carica* fiber by Scanning
499 Electron Microscope in Adsorption Isotherms studies of dye removal from aqueous
500 solution. *International Journal of Theoretical & Applied Science*, 4, 36.
- 501 Pathania, D., & Sharma, S. (2012) Synthesis and characterization of graft copolymers of
502 methacrylic acid onto gelatinized potato starch using chromic acid initiator in
503 presence of air. *Advance Material Letters*, 3, 136-142.
- 504 Pathania, D., & Sharma, S. (2012). Effect of surfactants and electrolyte on removal and
505 recovery of basic dye by using *Ficus carica* cellulosic fibers as biosorbent. *Journal of*
506 *Tenside Surfactants Detergents*, 49, 306-314.
- 507 Pathania, D., Sharma, S., & Singh, P. (2013). Removal of methylene blue by adsorption onto
508 activated carbon developed from *Ficus carica* bast. *Arabian Journal of Chemistry*,
509 <http://dx.doi.org/10.1016/j.arabjc.2013.04.021>.

- 510 Rathore, B. S., Gupta, V. K., Sharma, G., & Pathania, D. (2014). Synthesis, characterization
511 and antibacterial activity of cellulose acetate-tin (IV) phosphate nanocomposite.
512 *Carbohydrate Polymers*, 103, 221-227.
- 513 Rowell, R. M., James, S. H., & Jeffrey, S. R. (2002). Characterization and factors effecting
514 fiber properties, In E. Frollini, A. L. Leao, L. H. C. Mattoso. *Natural polymers and*
515 *agrofibres based composites* (pp.115-127). San Carlos, Brazil.
- 516 Sanghavi, B. J., Kalambate, P. K., Karna, S. P., & Srivastava, A. K. (2014). Voltammetric
517 determination of sumatriptan based on a graphene/gold nanoparticles/Nafion composite
518 modified glassy carbon electrode. *Talanta*, 120, 1-9.
- 519 Sanghavi, B. J., Mobina, S. M., Mathur, P., Lahiri, G. K., & Srivastava, A. K. (2013).
520 Biomimetic sensor for certain catecholamines employing copper(II) complex and silver
521 nanoparticle modified glassy carbon paste electrode. *Biosensors and Bioelectronics*, 39
522 (1), 124-132.
- 523 Sanghavi, B. J., Sitaula, S., Griep, M. H., Karna, S. P., Ali, M. F., & Swami, N. S. (2013).
524 Real-time electrochemical monitoring of adenosine triphosphate in the picomolar to
525 micromolar range using graphene-modified electrodes. *Analytical Chemistry*, 85 (17),
526 8158–8165.
- 527 Sanghavi, B. J., & Srivastava, A. K. (2011). Simultaneous voltammetric determination of
528 acetaminophen, aspirin and caffeine using an in situ surfactant-modified multiwalled
529 carbon nanotube paste electrode. *Electrochimica Acta*, 55, 8638-8648.
- 530 Sanghavi, B. J., & Srivastava, A. K. (2013). Adsorptive stripping voltammetric determination
531 of imipramine, trimipramine and desipramine employing titanium dioxide nanoparticles
532 and an Amberlite XAD-2 modified glassy carbon paste electrode. *Analyst*, 138, 1395-
533 1404.
- 534 Sharma, S., Pathania, D., & Singh, P. (2013). Preparation, Characterization and Cr (VI)
535 adsorption behavior study of poly (acrylic acid) grafted Ficus carica bast fiber.
536 *Advanced Materials Letters*, 4, 271-276.
- 537 Singh, V., Tiwari, A., Tripathi, D. N., & Sanghi, R. (2004). Microwave assisted synthesis of
538 guar-g-poly(acrylamide). *Carbohydrate Polymer*, 58, 1-6.
- 539 Singha, A. S., & Rana, R. K. (2010). Graft copolymerization of Methyl Methacrylate (MMA)
540 onto Agave Americana Fibers and Evaluation of their Physicochemical Properties.
541 *International Journal of Polymer Analysis and Characterization*, 15(1), 27-42.
- 542 Taylor, N. W., Fanta, G. F., Doane, W. M., & Russell, C. R. (1978). Swelling and rheology
543 of saponified starch-g-polyacrylonitrile copolymers. Effect of starch granule

- 544 pretreatment and grafted chain length. *Journal of Applied Polymer Science*, 22, 1343–
545 1357.
- 546 Tiwari, A., & Singh, V. (2008). Microwave-induced synthesis of electrical conducting gum
547 acacia-graft-polyaniline. *Carbohydrate Polymers*, 74, 427-434.
- 548 Wan, Z., Xiong, Z., Ren, H., Huang, Y., Liu, H., Xiong, H., Wu, Y., & Han, J. (2011). Graft
549 copolymerization of methyl methacrylate onto bamboo cellulose under microwave
550 irradiation. *Carbohydrate Polymers*, 83, 264-269.
- 551 Wang, L., Dong, W., & XU, Y. (2006). Synthesis and characterization of hydroxypropyl
552 methylcellulose and ethyl acrylate graft copolymers. *Carbohydrate Polymers*, 68, 626-
553 636.
- 554 Wang, S., & Zhu, Z. (2007). Effects of acidic treatment of activated carbons on dye
555 adsorption. *Dyes and Pigments*, 75, 306-314.
- 556 Zhenping, C., Xiulin, Z., Mingchen, J., Chen, L., & Zhang, L. (2003). Atom transfer radical
557 polymerization of methyl methacrylate with low concentration of initiating system
558 under microwave irradiation. *Polymer*, 44, 2243-2246.

Figure captions

Fig.1.(a-b) X-ray diffraction pattern of LcF and Lc-g-poly(MA/AAm) and (c-d) FTIR spectra of LcF and Lc-g-poly(MA/AAm) FTIR spectra of (a) LcF (b) Lc-g-poly(MA/AAm)

Fig.2.TGA spectra of (a) LcF (b) Lc-g-poly(MA/AAm)

Fig.3.SEM image of (a) LcF (b) Lc-g-poly(MA/AAm)

Fig.4.Percentage swelling onto (a) LcF (b) Lc-g-poly(MA/AAm) in different solvent (c) Effect of concentration on dye adsorption onto Lc-g-poly(MA/AAm) (d) Langmuir isotherm (e) Freundlich isotherm for dye adsorption onto Lc-g-poly(MA/AAm)

612

613

614

615

616

617

618

619

620 Table Captions

621

622 Table 1. Optimization of MA, AAm concentration and microwave exposure time

623 Table 2. Percentage crystallinity and crystallization index of LcF and Lc-g-poly(MA/AAm)

624 Table 3. Moisture absorbance, water uptake and chemical studies of LcF and Lc-g-
625 poly(MA/AAm)626 Table 4. Expressions for isotherm models, linearized forms, isotherms constants and
627 correlation coefficients for the adsorption of CR dye onto Lc-g-poly(MA/AAm)

628

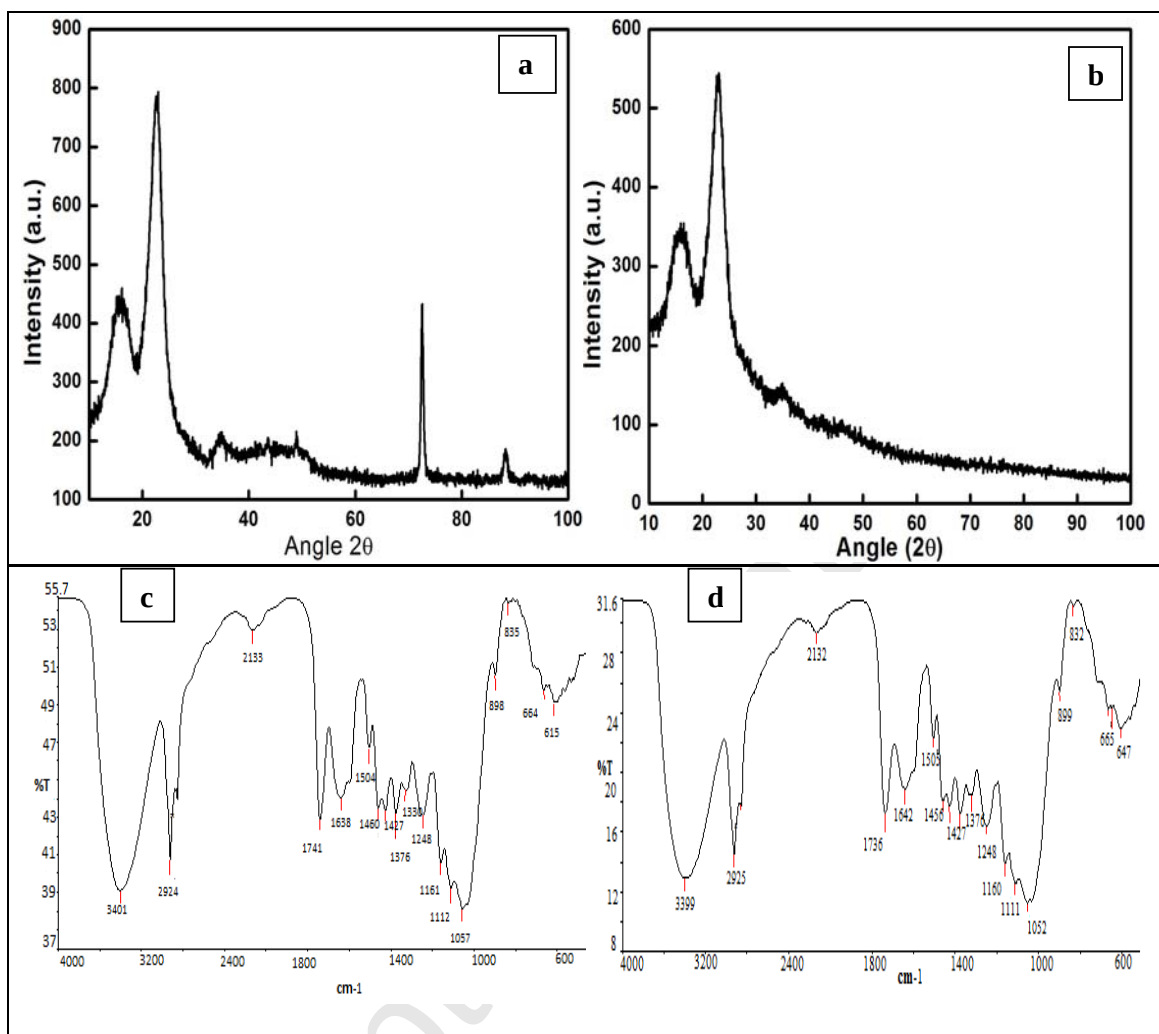


Fig.1. (a-b) X-ray diffraction pattern of LcF and Lc-g-poly(MA/AAm) and (c-d) FTIR spectra of LcF and Lc-g-poly(MA/AAm)

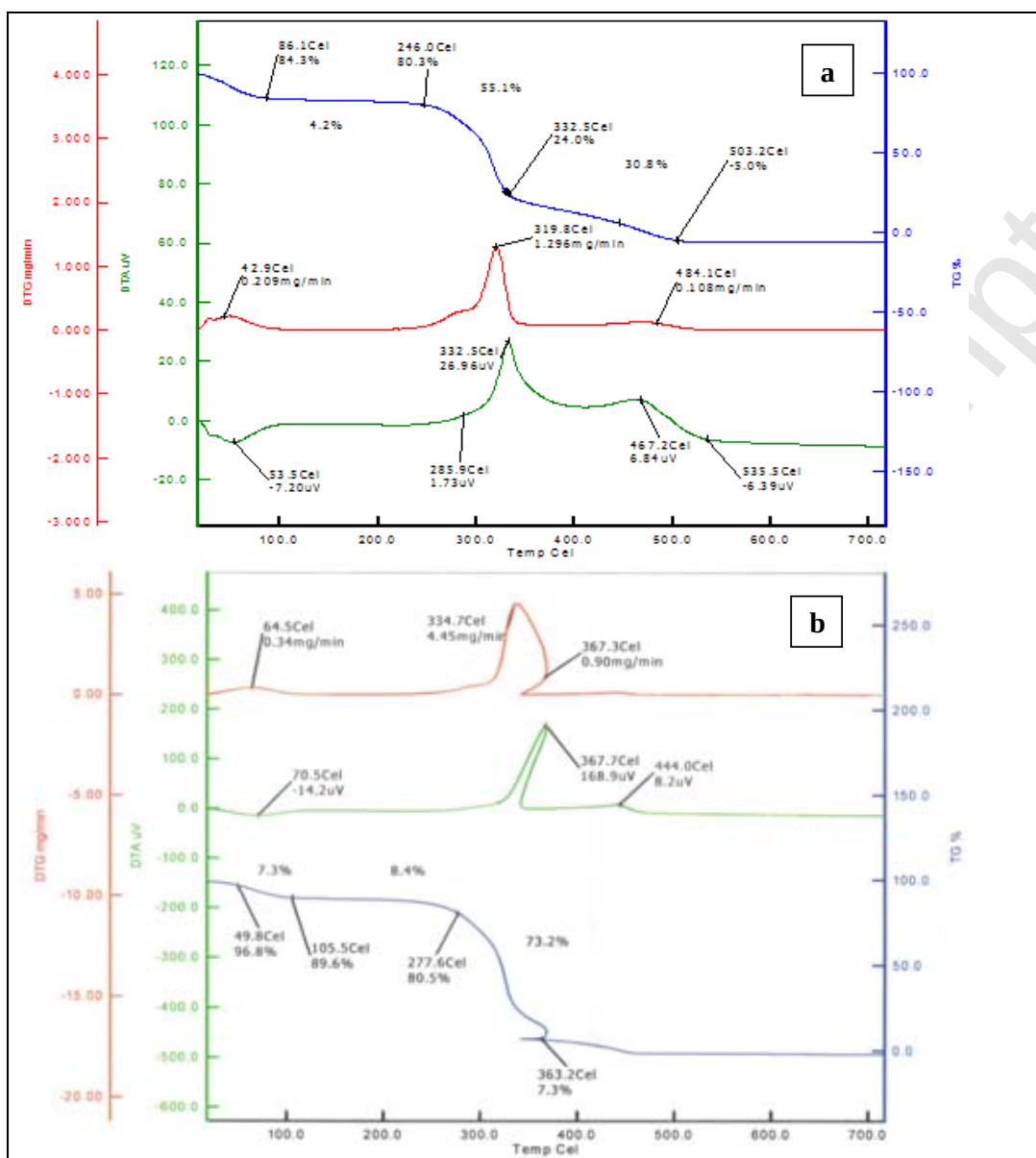


Fig.2. TGA spectra of (a) *LcF* (b) *Lc-g-poly(MA/AAm)*

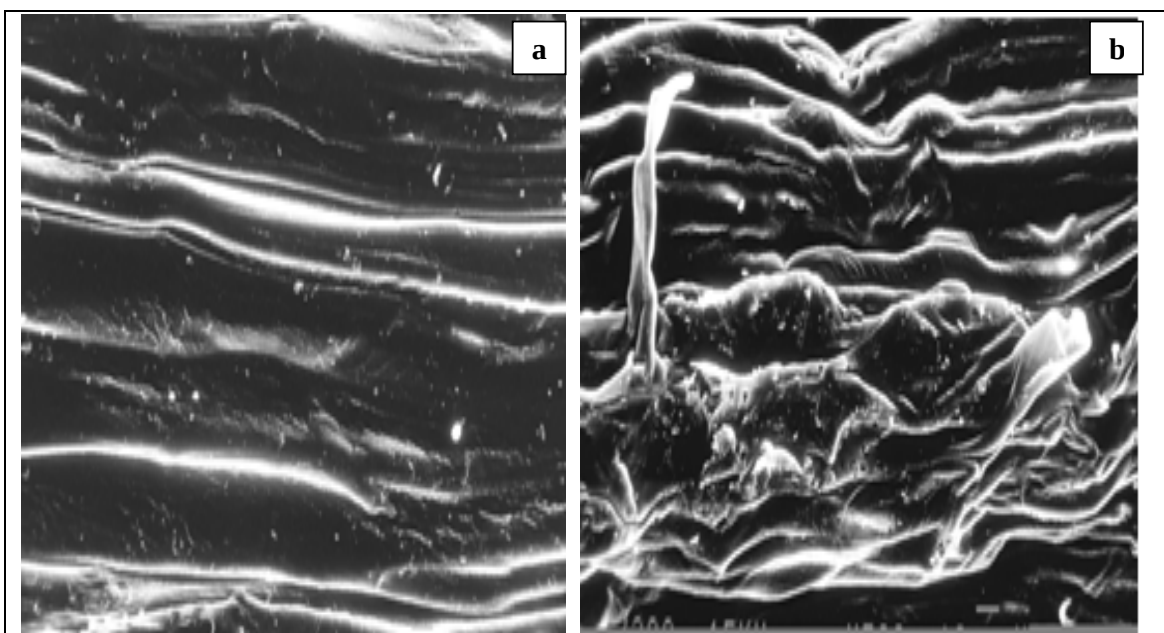


Fig.3. SEM image of (a) *LcF* (b) *Lc-g-poly(MA/AAm)*

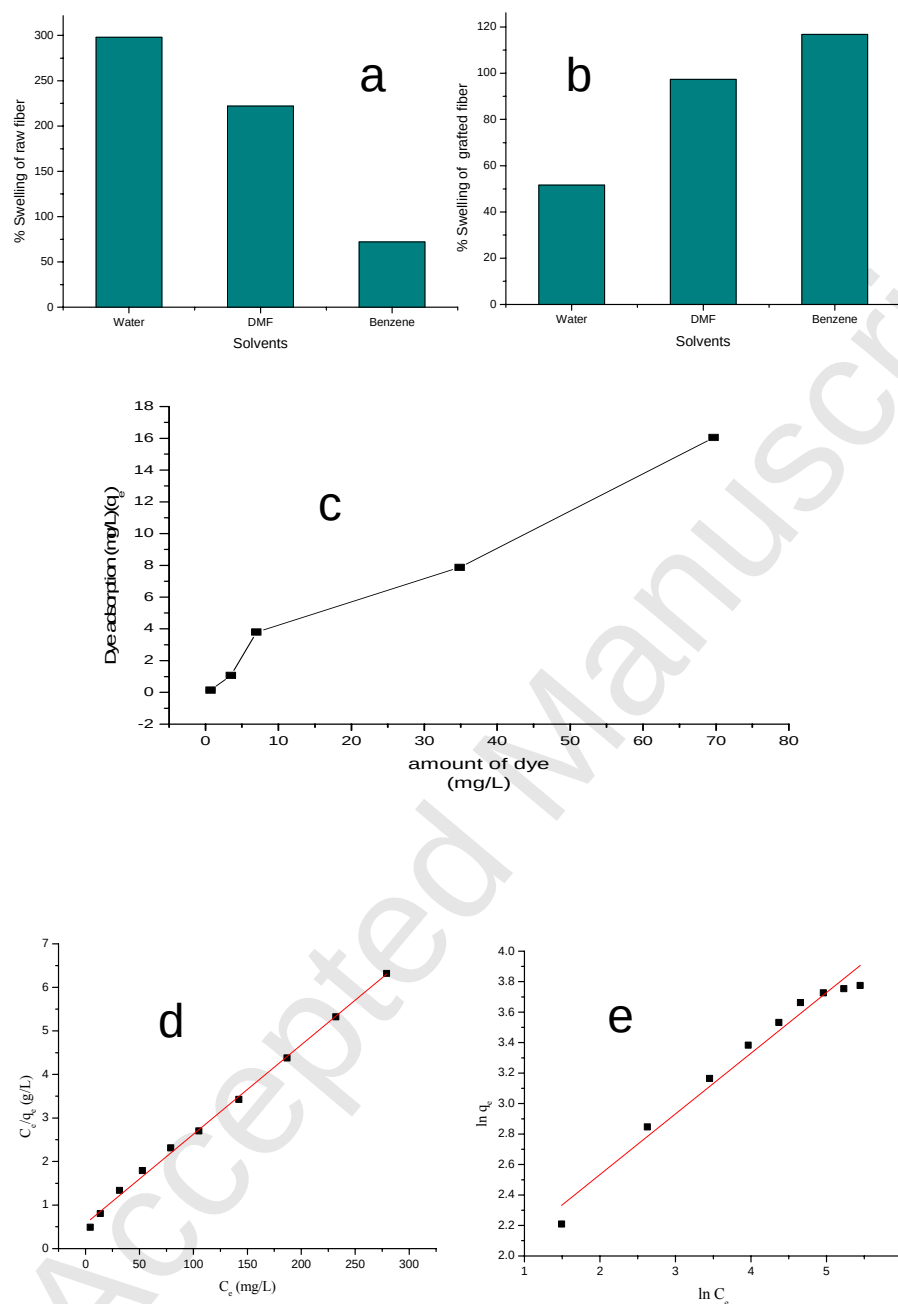
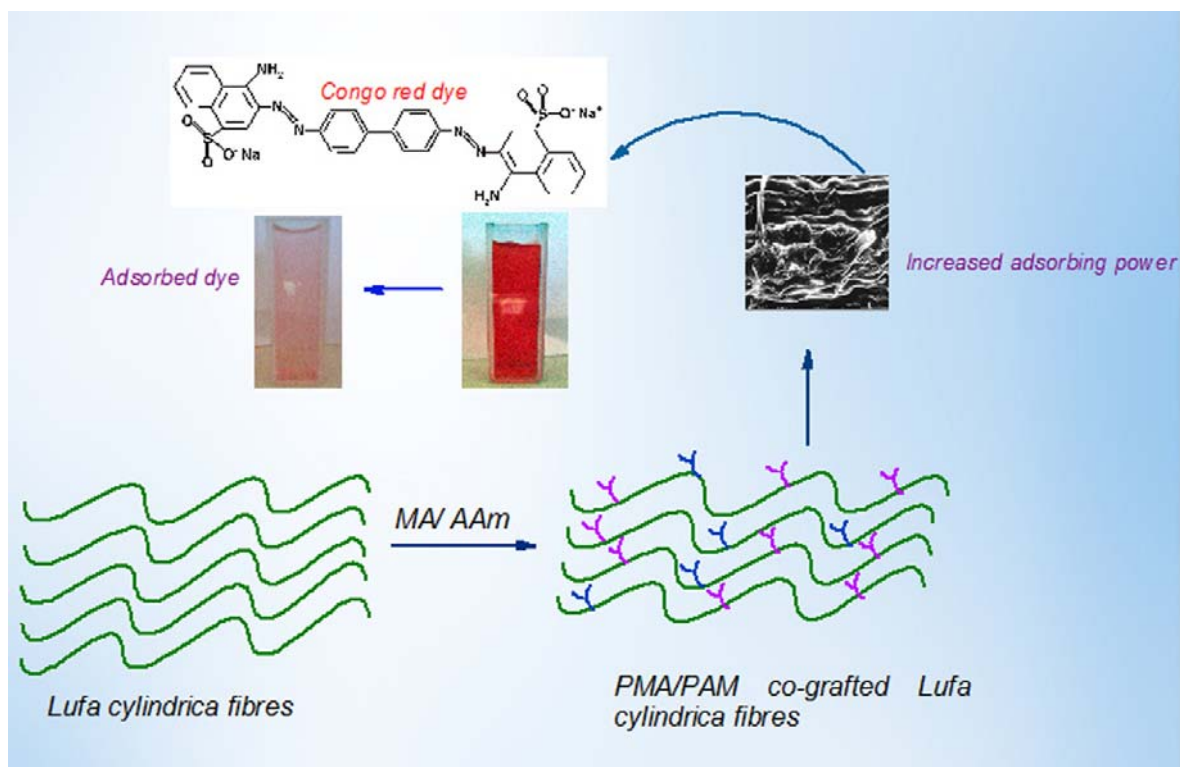


Fig.4. Percentage swelling onto (a) *LcF* (b) *Lc-g-poly(MA/AAm)* in different solvent (c) Effect of concentration on dye adsorption onto *Lc-g-poly(MA/AAm)* (d) Langmuir isotherm (e) Freundlich isotherm for dye adsorption onto *Lc-g-poly(MA/AAm)*

656



657
658
659

Table 1

Optimization of MA, AAm concentration and microwave exposure time

S.No.	MA/AAm ratio ($\times 10^{-3}$ mol/L)	% Grafting	MA/AAm ratio ($\times 10^{-3}$ mol/L)	% Grafting	Time (min)	% Grafting
1.	1.12 : 2.81	49.0	2.23 : 1.40	30.21	1	43
2.	2.23 : 2.81	48.1	2.23 : 2.81	51.08	2	48
3.	3.36 : 2.81	47.8	2.23 : 4.21	29.22	3	38
4.	4.48 : 2.81	43.6	2.23 : 5.62	27.83	4	24
5.	5.60 : 2.81	39.8	2.23 : 7.03	25.74	5	21

Table 2Percentage crystallinity and crystallization index of *LcF* and *Lc-g-poly(MA/AAm)*

Sample	2 θ (Deg)		Intensity		%Xc	C.I.
	Crystalline peak	Amorphous peak	I _c	I _A		
LcF	22.35	15.48	779	442	63.80	0.43
Lc-g-poly(MA/AAm)	24.23	16.40	546	352	60.80	0.35

Table 3Moisture absorbance, water uptake and chemical studies of *LcF* and *Lc-g-poly(MA/AAm)*

(a) Moisture absorbance

Sample	Initial weight of sample (g)	Final weight of sample (g)	%M _{abs}
<i>LcF</i>	0.5	0.74	48
<i>Lc-g-poly(MA/AAm)</i>	0.5	0.67	34

(b) Water uptake

Sample	Length of fiber wick (cm)	Water uptake (cm)
<i>LcF</i>	5.0	3.6
<i>Lc-g-poly(MA/AAm)</i>	5.0	2.5

(c) Chemical resistance

Sample	Percentage wt. loss in HNO ₃	Percentage wt. loss in NaOH
<i>LcF</i>	89.2%	74%
<i>Lc-g-poly(MA/AAm)</i>	67%	48%

711 **Table 4**

712 Expressions for isotherm models, linearized forms, isotherms constants and correlation

713 coefficients for the adsorption of CR dye onto *Lc-g-poly(MA/AAm)*

714

Isotherm	Expression	Linearized form	Parameters	Isotherm constants			
Langmuir	$q_e = \frac{q_m K_L C_e}{1 + K_L C_e}$	$\frac{1}{q_e} = \frac{1}{q_m} + \frac{1}{K_L q_m C_e}$	K_L, q_m	q_m (mg/g)	K_L (L/mg)	R_L	R^2
Freundlich	$q_e = K_F C_e^{1/n}$	$\ln q_e = \ln K_F + \frac{1}{n} \ln C_e$	K_F, n	17.39	0.021	0.427	0.995
				n	K_F (mg/g)	---	R^2
				2.63	4.49		0.918

715

716

717

718

719

719 Highlights
720 Luffa cylindrica fiber has been successfully grafted under microwave radiations.
721 Different properties of the fiber before and after grafting were investigated.
722 Fibers were characterized by different techniques
723 The grafted Luffa cylindrical fibers successfully removed congo red from water.
724
725